SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science &Engineering

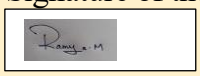
ASSESSMENT TOOL 2 – TECHNICAL PRESENTATION

Course Code	CSA0315	Course Title	Data Structures
CO Assessed	CO2	Assessment	Assessment Tool 2: Technical presentation
Weightage	30% of CIA	Total Marks	100
CO2 Statement	Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity. (BL3, BL4)		
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Section/Batch	Slot D / 2026	Date	08-10-2026

Code of Conduct:

I,Ramya M(1921260148), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

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CSA0315-Data Structure

File System Organization Using Trees

suggest a suitable non-linear data structure and justify the selection



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Submitted To:
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1. Problem Identification

Given Problem

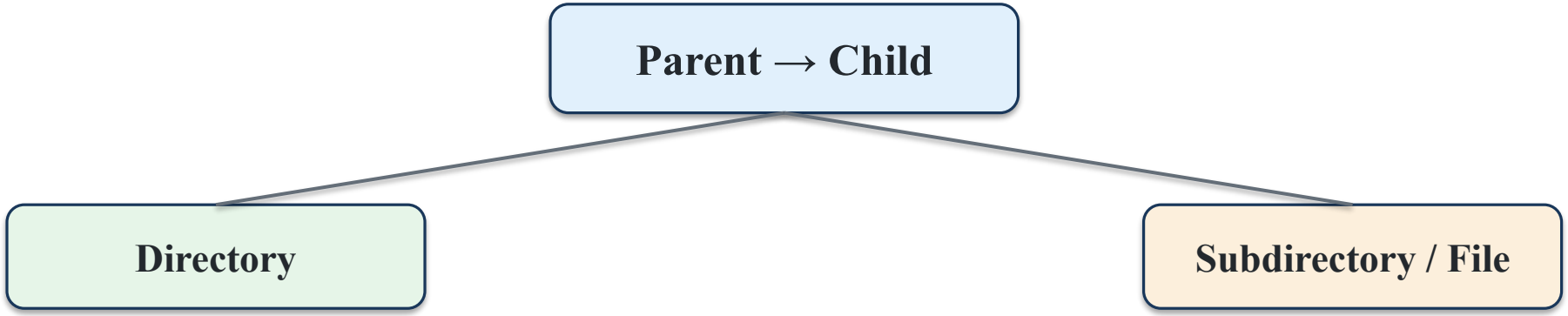
A file system contains folders, subfolders and files arranged at multiple levels.

Key Requirement

Represent parent–child relationships clearly and support traversal from the root to any file.

Why Linear DS Fails

A simple array/list does not naturally represent multiple parent–child levels.



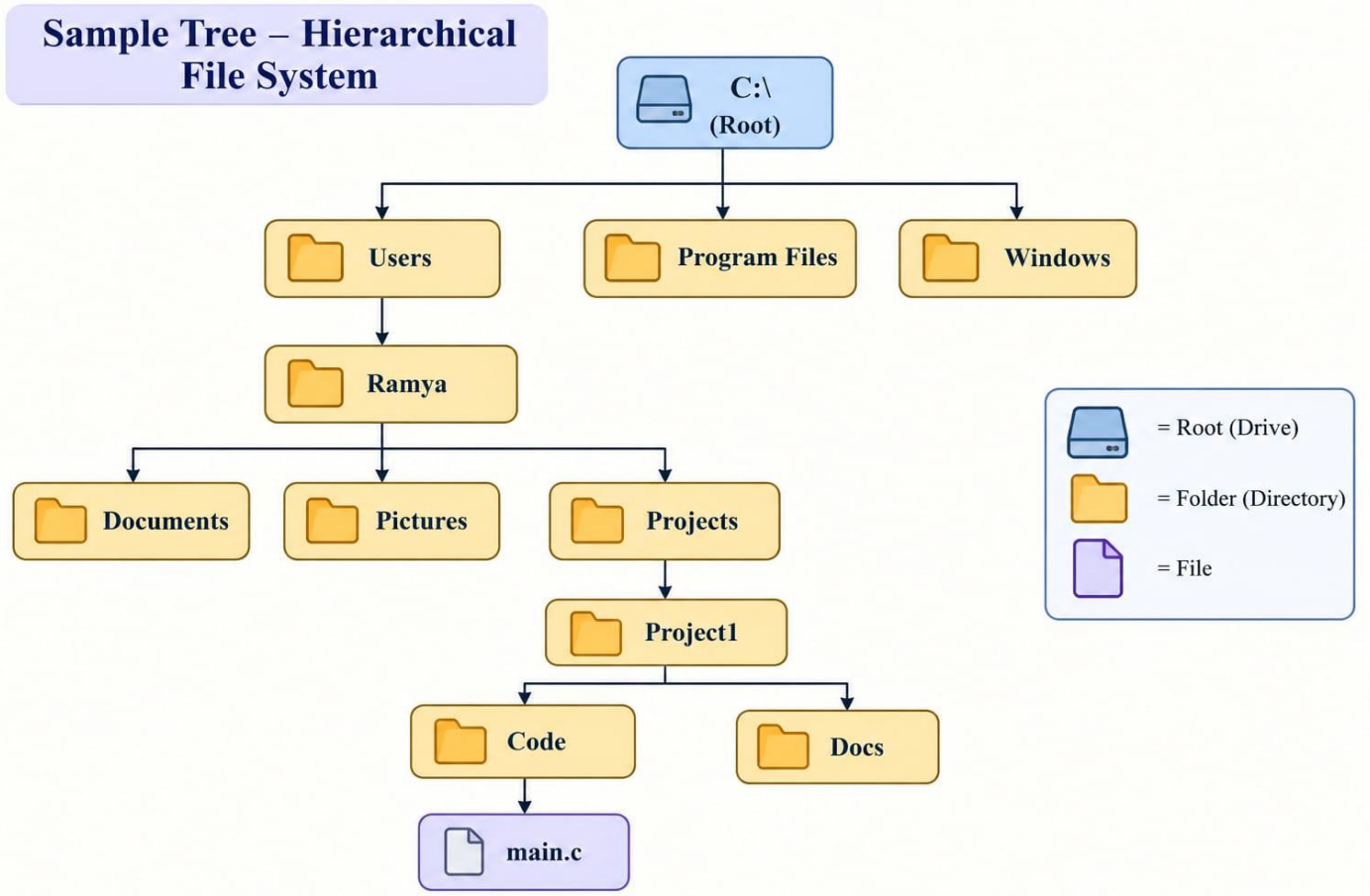
2. Recommended Data Structure: Tree

Tree ADT

A non-linear data structure made of nodes connected by edges. One node acts as the root and each child belongs to a parent.

File-System Mapping

Root → directories → subdirectories → files. Each directory can contain many children, while each child has one immediate parent.



3. Tree Representation

General Tree Node

Each node may store:
- Name / file metadata
- Parent reference
- Links to its children

This supports any number of subdirectories.

Common Implementation

For a general tree, children can be maintained using:
- Linked list of child pointers
- First-child / next-sibling representation

Both handle variable numbers of children.

```
Node {
    name;
    parent;
    children[];
}
```

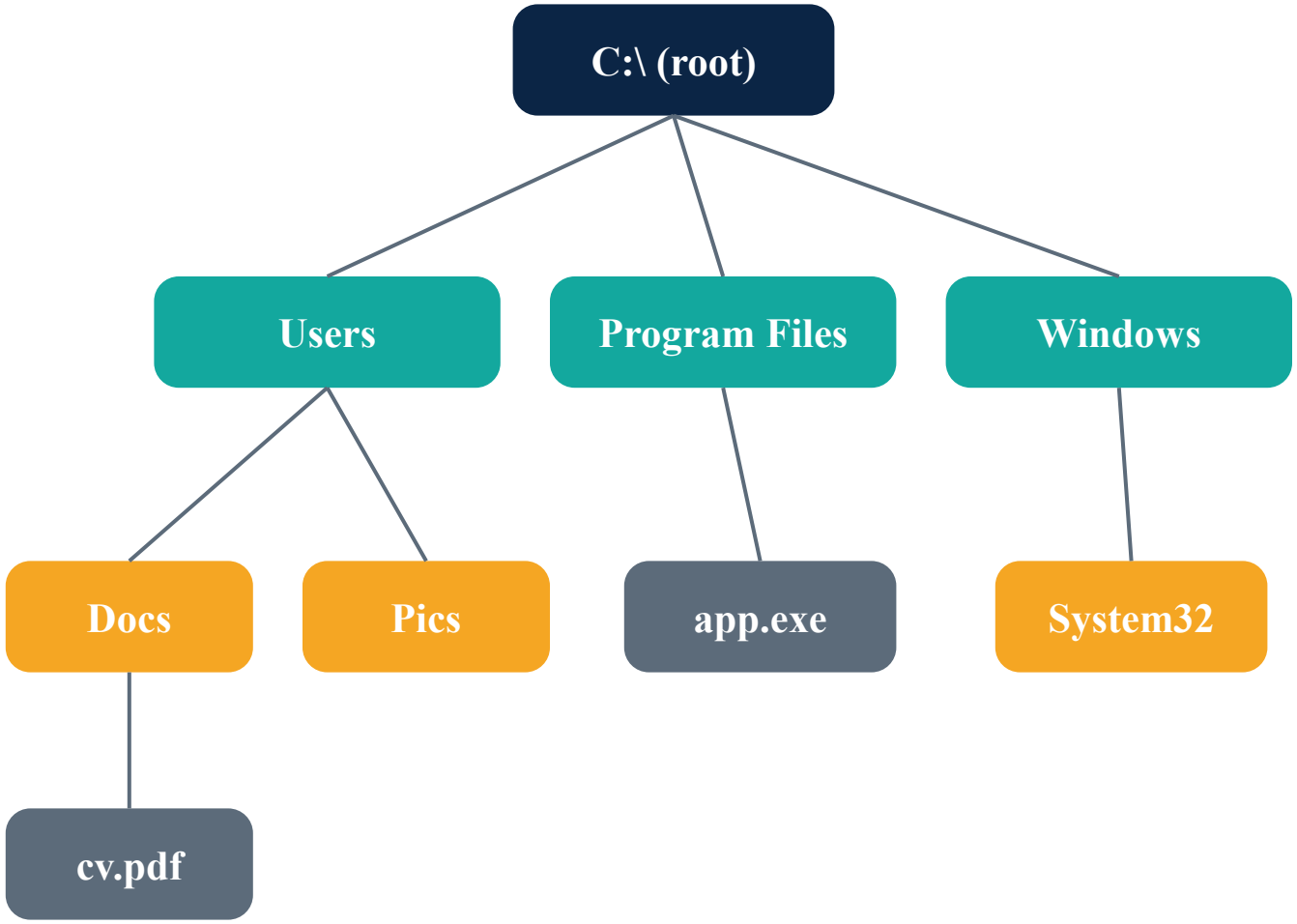
4. Operations & Complexity

Operation	Typical complexity	Reason
Insert child	O(1)*	If parent node is already known and child is linked directly.
Delete subtree	O(k)	k = number of nodes removed from that subtree.
Search	O(n)	General tree may require traversal of nodes.
Traverse	O(n)	Each reachable node is visited once.

Important

*O(1) insertion assumes the parent is already located. If the parent must first be searched, the search cost must also be included.

5. Tree Model Of A File System



Terminology

- Root: top directory
- Internal node: folder
- Leaf node: file / empty folder
- Edge: parent-child link
- Path: root to node = unique address
- Depth: nesting level

6. Why Tree Is the Best Choice

Tree ✓

- Directly represents hierarchy
- Natural parent–child relationship
- Supports recursive traversal
- Scales to many directory levels
- Easy to represent paths
- Suitable for DFS/BFS traversal

Array / List

- Mainly linear
- Hierarchy is not natural
- Insert/delete may require shifting or extra indexing
- Relationships need additional bookkeeping

Graph

- More general than required
- Allows many-to-many relationships
- File systems normally need a simpler hierarchical model
- Tree is more direct and easier to manage

Conclusion: Use a general tree because the problem itself is hierarchical.

7. Conclusion

1. A file system is hierarchical, so a linear structure cannot represent it.
2. A Tree (n-ary tree) is the most suitable non-linear data structure.
3. Folders act as internal nodes and files as leaf nodes.
4. Every item has one parent, giving a unique path from the root.
5. Search, insert and delete take O(h), far faster than O(n) linear scan.
6. Recursive traversal (DFS / BFS) makes listing and managing files simple.
7. Real operating systems such as NTFS, ext4 and APFS use tree-based directories.

“Hierarchy is best represented by a tree.”